BANGLADESH UNIVERSITY OF BUSINESS & TECHNOLOGY
(BUBT)

CSE 326: Micro Processor & Micro Controller Lab Project

SUBMITTED BY

Name	ID	Intake
Al Adid	20245103007	53
Md Fahim Islam	20245103035	53
Md Mahfuzul Bashar Shaikot	20245103038	53
Md. Sakibur Rahman	20245103091	53
Abu Ahmad Mahfuzor Rahman	20234103092	53

SUPERVISED BY:

Fahima Khanam
Assistant Professor,
Department of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY
(BUBT)

16, July, 2026

Project Name : ArcticGuard

Objective :

The Smart AC Monitoring and Fire Prevention System is an Arduino Uno-powered safety and monitoring system that continuously monitors the status of the air conditioner's operation. The system is capable of detecting any early indications of trouble such as gas leakage or smoke, high surface temperature, and unusual vibration, which are often associated with problems with the functioning, maintenance, and/or electric components of the machine. By means of the timely notification system, one can determine whether an air conditioner requires cleaning, maintenance, or fixing at an early stage.

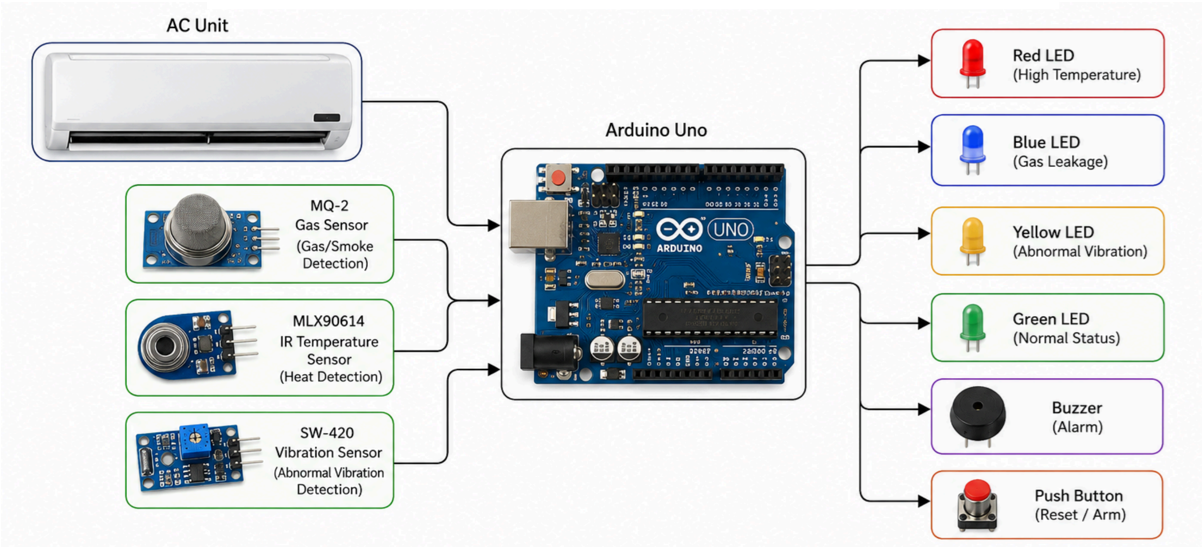
Functions :

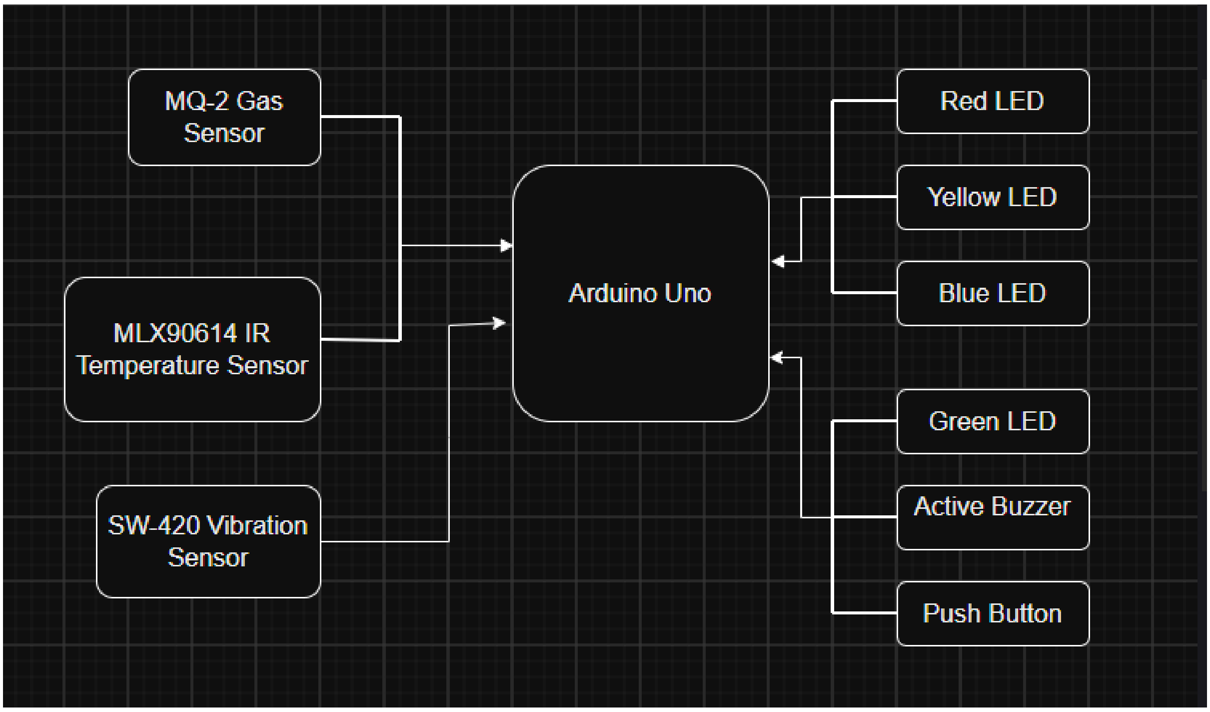
- Detect gas leakage using an MQ-2 gas sensor.
- Monitor excessive heat using an IR temperature sensor (MLX90614).
- Detect abnormal vibration using a SW-420 vibration sensor.
- Analyze sensor data using Arduino Uno.
- Provide visual alerts using dedicated LEDs.
- Activate a buzzer when abnormal conditions are detected.
- Reset or arm the system using a push button.

Required equipment (hardware) :

- Arduino Uno
- MQ-2 Gas Sensor
- MLX90614 IR Temperature Sensor
- SW-420 Vibration Sensor
- Red LED (High Temperature Alert)
- Yellow LED (Abnormal Vibration Alert)
- Blue LED (Gas Leak Alert)
- Green LED (System Normal)
- Active Buzzer
- Push Button (Reset/Arm)
- Breadboard or PCB
- 220Ω Resistors
- Jumper Wires (As Required)
- 5V Power Supply / USB Adapter

Project Design :





Working Principle :

1. The MQ-2 gas sensor continuously monitors the air around the air conditioner for smoke or combustible gases that may indicate overheating or an electrical fault.

2. The MLX90614 infrared temperature sensor continuously measures the surface temperature of the AC unit without physical contact to detect abnormal overheating.
3. The SW-420 vibration sensor monitors the vibration of the compressor and fan to detect unusual vibration caused by mechanical faults or wear.
4. The Arduino Uno continuously collects and processes data from all three sensors.
5. The sensor readings are compared with predefined safety threshold values to determine whether the AC is operating normally or showing signs of a fault.
6. During normal operation, the green LED remains ON, indicating that the AC is functioning properly.
7. If abnormal vibration is detected, the yellow LED turns ON and the buzzer sounds to indicate that the AC may require inspection or maintenance.
8. If the AC surface temperature exceeds the safe limit, the red LED turns ON and the buzzer alerts the user, indicating possible overheating.
9. If smoke or combustible gas is detected, the blue LED turns ON and the buzzer sounds continuously to warn of a potential fire hazard.
10. If two or more abnormal conditions occur at the same time, the corresponding LEDs illuminate together and the buzzer provides a continuous alarm, indicating a critical condition that requires immediate inspection or shutdown of the air conditioner.
11. The push button is used to reset the alarm or re-arm the monitoring system after the issue has been inspected and resolved.

Project Application :

- **Residential Air Conditioners:** Monitors home air conditioning systems to detect overheating, gas leakage, or abnormal vibration, helping homeowners identify potential problems before they become hazardous.
- **Office Buildings:** Ensures the safe and reliable operation of office AC systems by providing early fault detection, reducing unexpected breakdowns and maintaining a comfortable working environment.
- **Commercial Buildings:** Suitable for shopping malls, restaurants, banks, and other commercial facilities where continuous air conditioning is essential for customer comfort and equipment protection.
- **Hospitals and Healthcare Facilities:** Helps maintain uninterrupted operation of air conditioning systems in critical environments where temperature control is important for patient care and medical equipment.
- **Hotels:** Monitors air conditioners in guest rooms and common areas, reducing maintenance costs and improving guest comfort by detecting faults before they cause failures.

- **Educational Institutions:** Can be installed in classrooms, laboratories, libraries, and computer labs to ensure safe and efficient operation of air conditioning systems.
- **Laboratories:** Protects laboratories by detecting abnormal AC conditions that could affect sensitive experiments or equipment due to overheating or system failure.
- **Data Centers and Server Rooms:** Helps prevent overheating and equipment damage by monitoring the air conditioning systems responsible for cooling servers and networking devices.
- **Industrial Facilities:** Monitors industrial air conditioning units used in factories and production areas, helping reduce downtime caused by unexpected AC failures.
- **Maintenance and Service Centers:** Can be used as a demonstration or diagnostic tool for technicians to identify common warning signs before performing maintenance or repairs.

Advantages:

- **Early Fault Detection:** Continuously monitors key indicators such as temperature, gas/smoke, and vibration to identify developing problems before they become critical.
- **Fire Risk Reduction:** Detects overheating and smoke at an early stage, helping reduce the possibility of AC-related fires and improving overall safety.
- **Predictive Maintenance:** Alerts users when the air conditioner may require cleaning, servicing, or repair, preventing minor issues from developing into major failures.
- **Improved Equipment Reliability:** Continuous monitoring helps maintain stable AC performance and reduces unexpected system breakdowns.
- **Real-Time Monitoring:** The Arduino Uno processes sensor data continuously, allowing immediate detection of abnormal operating conditions.
- **Instant Visual and Audible Alerts:** Dedicated LEDs indicate the type of fault, while the buzzer immediately notifies users whenever dangerous conditions are detected.
- **Low-Cost Implementation:** Built using affordable and widely available Arduino components.
- **Easy Installation and Maintenance:** The system requires minimal hardware, making it simple to install, troubleshoot, and maintain.
- **Energy-Efficient Operation:** The monitoring system consumes very little power while operating continuously, making it suitable for long-term use.
- **Extendable Design:** Additional sensors, wireless communication modules, or IoT features can easily be integrated in the future for remote monitoring and advanced diagnostics.

- **Increases Air Conditioner Lifespan:** By identifying issues early and encouraging timely maintenance, the system helps reduce wear and tear, extending the service life of the air conditioner.
- **Improves User Safety:** Provides users with sufficient warning to inspect or shut down the air conditioner before a hazardous condition develops, helping protect both people and property.